

AFREP Aircraft / Drone Technician

1. Position Overview

The AFREP Aircraft / Drone Technician provides technician-level maintenance, troubleshooting, wiring repair, and electronic repair support for unmanned aerial systems (UAS/drones), associated interconnects, and avionics or electronic components in support of the 355th Maintenance Group (MXG) Aircraft Repair Enhancement Program (AFREP) at Davis-Monthan Air Force Base.

Contractor personnel perform hands-on technical tasks in accordance with approved technical data, workmanship standards, and safety requirements while operating under Government direction and Quality Assurance (QA) oversight. This position augments organic AFREP capability to meet increased mission workload and sustain continuity of operations. The contractor does not exercise supervisory authority, approve maintenance actions, certify airworthiness, or perform inherently governmental functions.

2. Prerequisites

The contractor shall possess prior technical experience directly related to aircraft electronics, avionics, unmanned systems, or comparable Department of Defense (DoD) maintenance environments. The contractor must be capable of working in a regulated maintenance setting and complying with Air Force technical, safety, security, and configuration control requirements.

3. Tasks

The contractor shall perform the following tasks, including but not limited to:

3.1 Unmanned Systems Maintenance and Repair

Perform maintenance, inspection, troubleshooting, and repair of unmanned aerial vehicles (UAVs/drones) and associated subsystems in accordance with approved technical data and AFREP work instructions.

Execute electronic and mechanical repair actions including component replacement, wiring repair, soldering, connector installation, and system reassembly.

Conduct functional checks and system verification following maintenance actions.

3.2 Wiring, Interconnect, and Connector Repair

Perform basic and intermediate wire repair in support of unmanned systems and avionics maintenance, including wire splicing, crimping, pin replacement, and connector repair, in accordance with approved technical data and workmanship standards.

Accomplish environmental and non-environmental splice repairs, including butt splices, lap splices, and solder sleeve repairs, as applicable.

Repair, replace, and re-terminate electrical connectors, including but not limited to:

Circular Connectors (Canon-type / MIL-DTL-38999 series)

Micro-D connectors (MIL-DTL-83513)

D-Subminiature connectors

Miniature and board-level connectors commonly used in unmanned systems, such as JST, Molex, Hirose, and equivalent connector families

Perform pin and socket removal and installation, wire stripping, crimping, and continuity verification using approved tools.

Inspect wiring and connectors for defects such as broken conductors, insulation damage, corrosion, bent or recessed pins, improper strain relief, or poor workmanship.

Verify wiring integrity through continuity checks, insulation resistance checks (as required), and functional testing following repair actions.

Apply proper shielding, grounding, bonding, and strain-relief techniques appropriate to aircraft and unmanned systems environments.

Comply with tool control and foreign object damage (FOD) prevention requirements during all wiring and connector repair activities.

3.3. Avionics and Electronics Support

Provide avionics and electronics maintenance support for manned aircraft components as required and within AFREP scope.

Troubleshoot and isolate faults using technical orders, schematics, and electronic test equipment.

Assist Government personnel with repair actions during periods of elevated workload.

3.4 Test equipment and Tool Utilization

Use and maintain hand tools and electronic test equipment including multimeters, oscilloscopes, power supplies, and automated or semi-automated test equipment.

Comply with calibration, tool accountability, and equipment serviceability requirements.

Report tool, test equipment, or calibration discrepancies to designated Government personnel.

3.5 Documentation and Maintenance Records

Prepare and update maintenance documentation, repair records, and electronic tracking entries as required by AFREP processes.

Ensure documentation accuracy, completeness, and traceability in accordance with QA standards.

Identify, document, and report workmanship defects, rework requirements, or recurring discrepancies to Government personnel.

3.6 Configuration Control and Acceptance Authority

Perform all maintenance and repair actions strictly in accordance with approved technical data and Government-directed configurations.

The contractor shall not deviate from approved configurations or repair processes without prior Government direction or approval.

The contractor shall not certify airworthiness, approve maintenance action, sign maintenance releases, or accept repaired components on behalf of the Government.

3.7 Test Flights (Limited)

Perform occasional unmanned system test flights only when:

Directed by the Governments,

Proper qualifications and training are documented,

A minimum Secret security clearance is maintained, and

All applicable FAA, DoD, and installation requirements are met.

The contractor shall not plan, authorize, or approve flight operations.

3.8 Safety, Environmental, and Compliance

Follow all applicable Air Force, OSHA, and installation safety requirements.

Use required personal protective equipment (PPE).

Handle, store, and dispose of hazardous materials (e.g., solder, flux, solvents, adhesives, batteries, and electronic waste) in accordance with applicable Air Force, OSHA, and environmental regulations.

Promptly report hazards, unsafe conditions, or procedural deviations to Government representatives.

3.9 Continuity of Operations

Support continuity of AFREP technical capability and repair operations during periods of workload surge, personnel turnover, and mission transition by maintaining consistent repair practices and documentation accuracy.

4. Qualifications

4.1 Knowledge

The contractor shall possess working knowledge of:

Aircraft and unmanned systems maintenance and repair practices

Electronic principles including circuitry, power distribution, signal flow, and component-level repair

Aircraft and unmanned systems wiring practices, including splicing, crimping, pinning, and connector repair

MIL-spec circular connectors (e.g., MIL-DTL-38999) and miniature connectors used in unmanned systems

Shielding, grounding, bonding, and strain-relief techniques

Use of electronic test equipment and interpretation of technical schematics

Tool control and QA-driven maintenance environments

Controlled Unclassified Information (CUI) handling requirements applicable to technical data and maintenance records.

4.2 Education

- High School diploma or GED is required.
- FAA Airframe & Powerplant (A&P) license, OEM, or DoD depot-level repair training (e.g., Raytheon, Lockheed Martin, Northrop Grumman courses) is preferred.

4.3 Training and Certifications

- The following IPC Certifications or equivalents are required:
 - IPC-7711/7721: Rework, modification, and repair of electronic assemblies
 - IPC-A-610: Acceptability of electronic assemblies
 - IPC-J-STD-001: Soldered electrical/electronic assemblies
- Additional AFREP specific training will be provided by AFREP or other agencies on DMAFB.

4.4 Military Standards Familiarity

- MIL-STD-883, MIL-STD-454, MIL-STD-1686, etc.
- Technical Order (TO) compliance and Air Force documentation standards
- Quality Assurance (QA) and FOD prevention training

4.5. Experience

- 3–10+ years of electronics or avionics repair experience
- Component-level troubleshooting (down to resistors, ICs, microchips)
- Ability to read schematics, wiring diagrams, and TOs
- Experience repairing aircraft LRUs, SRUs, or circuit cards
- Soldering and micro-soldering proficiency

5. Security Clearance and Access

Contractor personnel must have Secret security clearance to perform duties.

Must comply with REAL ID requirements and all base access policies.

6. Physical and Environmental Requirements

Ability to lift and carry items up to 40 pounds, heavier items with assistance.

Ability to stand, bend, crouch, reach, and perform fine motor tasks for extended periods.

Work is performed in maintenance shops, controlled areas, and aircraft support environments.

7. Other Qualifications

Ability to read, write, and speak English clearly.

Ability to operate computer systems and maintenance tracking tools.

Ability to multi-task in a production-driven environment.

This position is non-supervisory and does not include authority to direct Government personnel or make final technical or operational decisions.